STEAMPUNK SKYFORGE

MOCK FRC GAME · THEME EDITION

FIRST® Robotics Competition – style MOCK GAME · Steampunk theme edition

Theme edition: only the name, colours and logo differ. Field, scoring and rules are identical to the SKYLINE edition. The steampunk styling is a fan homage and is not affiliated with or endorsed by FRC Team 1577.

Game Manual (Field, Scoring, Game Rules, ROBOT size rules) – Version 0.1 draft – 2026-09-22

> **Unofficial.** This is a mock game written for design-practice purposes. It is not affiliated with or endorsed by *FIRST*®. Structure, penalty values, conduct rules and ROBOT construction rules are modelled on the 2025 *REEFSCAPE* Game Manual (with 2019 *DESTINATION: DEEP SPACE* used for the height-limit zone concept). Where this manual says "inherited", the rule of that number in the 2025 manual applies with game terms substituted (Section 7.0). Nothing in this document has been prototyped; all numbers are design proposals.

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1 Introduction & conventions

1.1 Units, tolerances and coordinates

- Dimensions are in inches with millimetres in parentheses. FIELD dimensions are nominal, ±1 in. (±25 mm) unless noted; teams should design for this variation, as in real FRC events.

- **Coordinate system** (used throughout, Figure 4-1): origin **O** is the carpet-level corner where the red ALLIANCE WALL meets the GOAL-SIDE FIELD BORDER. **+X** runs across the FIELD toward the FEEDER-SIDE border (0 → 324 in.). **+Y** runs along the FIELD toward the blue ALLIANCE WALL (0 → 648 in.). **Z** is height above the carpet.
- **Symmetry:** the blue end is the **mirror image** of the red end across the CENTER LINE (Y = 324 in.), with colours swapped: a red item at (X, Y) has a blue counterpart at (X, 648 − Y). It is a reflection, not a 180° rotation, so both alliances have their HORIZONTAL GOALS on the same GOAL-SIDE border (X = 0).
- ZONE dimensions are to the outside edge of the 2 in. (51 mm) boundary tape; the tape is part of the ZONE. ZONES are infinitely tall volumes.

1.2 Source concept

The layout below was translated from the designer's sketch. The mapping of sketch symbols to game elements is:

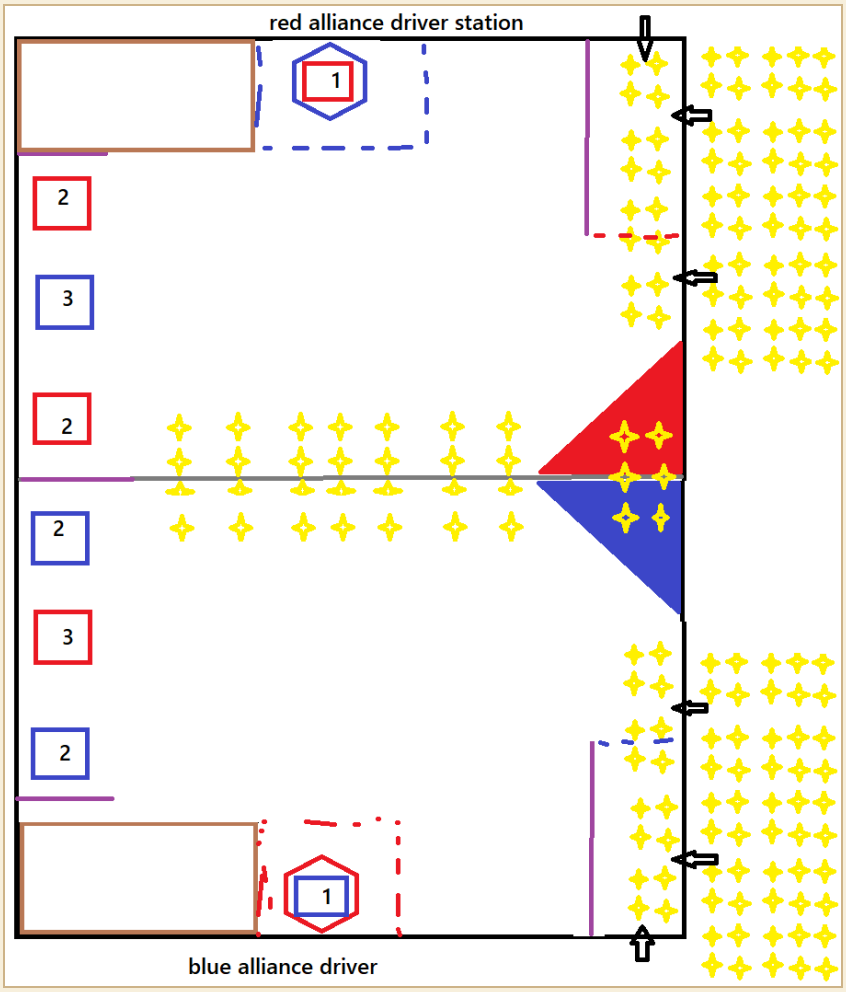


Figure 1-1 Source sketch (not to scale; red driver station at top).

Sketch symbol	Game element	Section

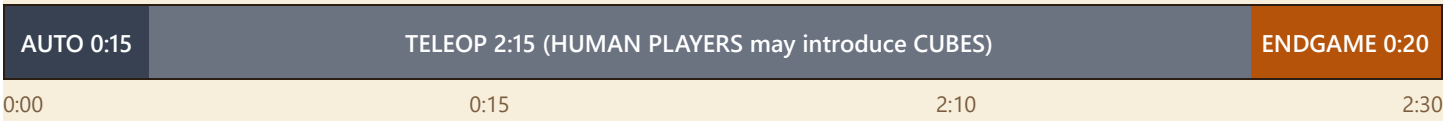
Brown rectangle	STARTING ZONE (1 per alliance, 3 ROBOTS, against own ALLIANCE WALL)	4.4
Purple lines	GUARDRAILS (internal barriers). The outer boundary is called the FIELD BORDER in this manual to avoid confusion.	4.3
Rectangles 1 / 2 / 3	HORIZONTAL GOALS – GOAL 1 (low), GOAL 2 (mid, ×2), GOAL 3 (high), per alliance	4.5
Hexagon	VERTICAL GOAL, mounted on top of the *opponent* ALLIANCE WALL; the opponent's GOAL 1 stands directly in front of it as a blocker	4.6
Dashed line around hexagon	LAUNCH ZONE (a ROBOT must intersect it to deliver into its VERTICAL GOAL)	4.6
Arrows	INTRODUCTION POINTS (IP-A, IP-B, IP-C) for the HUMAN PLAYER, TELEOP only	4.8
Solid triangles	FLOATING BOARDS (elevated plates; HANG on them, or drive UNDER if the ROBOT is short enough)	4.7
Dashed line closing the feeder lane	SAFE ZONE (the only protected zone)	4.4
Yellow stars	CUBES (staged on the FIELD and in reserve outside the FIELD)	4.9–4.10

2 Game overview

In **Steampunk SKYFORGE**, two ALLIANCES of three ROBOTS collect foam CUBES and score them in HORIZONTAL GOALS at three heights along the GOAL-SIDE wall and in a hexagonal VERTICAL GOAL mounted on top of the *opponent's* ALLIANCE WALL. HUMAN PLAYERS introduce new CUBES through three INTRODUCTION POINTS from TELEOP onward. At the end of the MATCH, ROBOTS HANG from their ALLIANCE's FLOATING BOARD or park fully UNDER it.

The central design trade-off is the **FLOATING BOARD clearance of 32 in. (813 mm)**. A short ROBOT, typically one that only places into GOAL 1 (18 in. rim), can drive under the boards. That gives it CUBES staged under the boards, a short path along the feeder side and UNDER points. Taller ROBOTS can reach GOAL 2, GOAL 3 and the VERTICAL GOAL but have to go around the boards. A ROBOT may reach at most two CUBE heights above the GOAL 3 rim (78 in. total height).

AUTO and TELEOP scoring have the same point values.



ENDGAME is the last 20 seconds of TELEOP. The total MATCH is 2:30, the same as 2025.

3 Glossary

Terms not listed here (e.g. BUMPER, BUMPER ZONE, CONTROL, MOMENTARY, CONTINUOUS, REPEATED, PIN, ROBOT PERIMETER, DRIVE TEAM, FIELD STAFF) have their 2025 meanings. In 2025, MOMENTARY means fewer than approximately 3 seconds and CONTINUOUS means more than approximately 10 seconds.

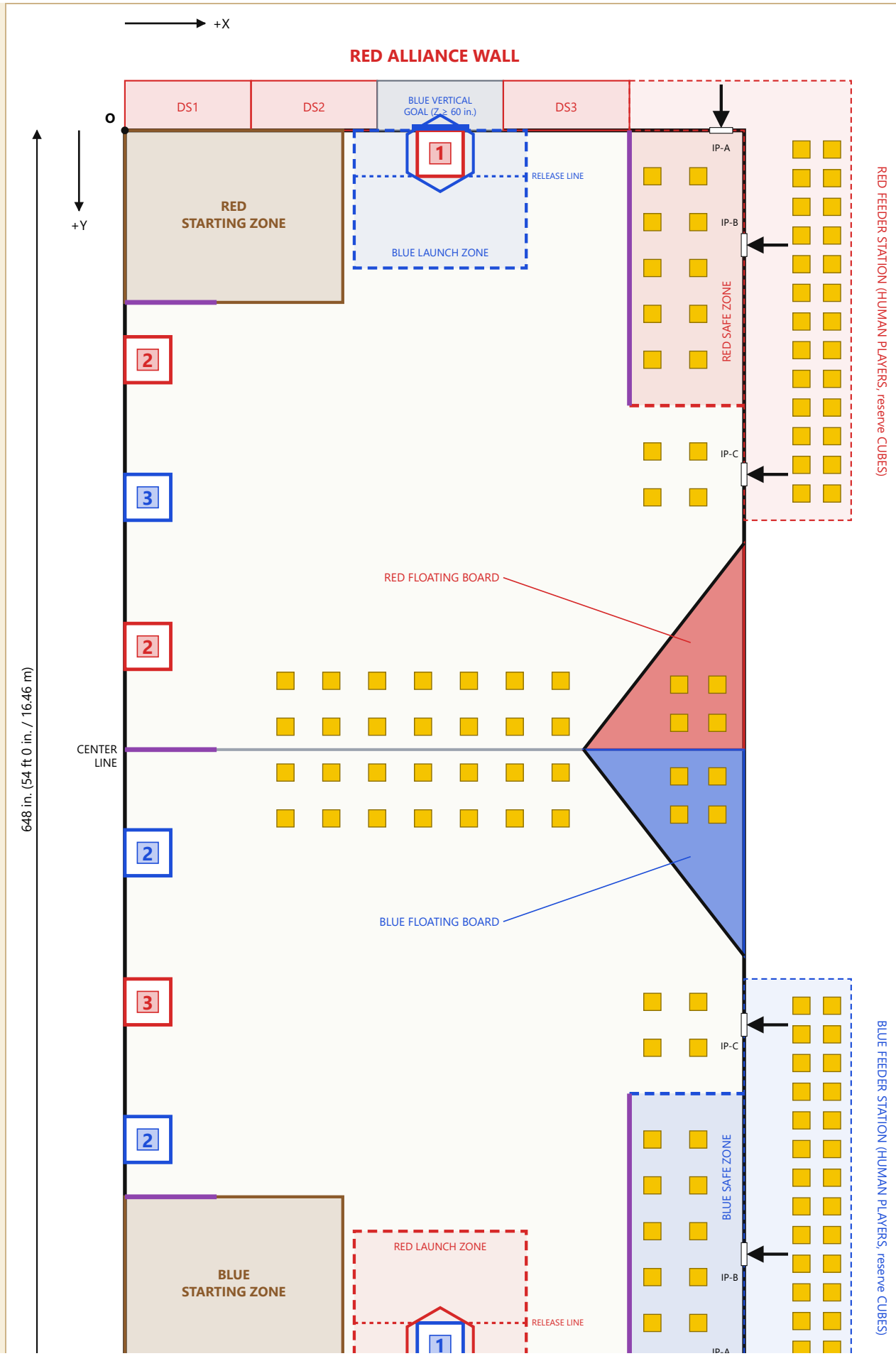
Term	Definition
ALLIANCE WALL	The FIELD boundary at each end (Y = 0 red, Y = 648 in. blue), comprising 3 DRIVER STATIONS, the VERTICAL GOAL section and the corner of the FEEDER STATION.
CENTER LINE	Grey tape line at Y = 324 in. dividing the FIELD into the red half (Y < 324) and the blue half.
CUBE	The SCORING ELEMENT: a 9 in. (229 mm) solid foam cube (Section 4.9).
ENDGAME	The final 20 seconds of TELEOP.
FEEDER STATION	The area outside the FEEDER-SIDE FIELD BORDER, from the ALLIANCE WALL to 204 in. from it, where the ALLIANCE's HUMAN PLAYERS and reserve CUBES are located.
FIELD BORDER	The 20 in. (508 mm) tall polycarbonate wall on the two long sides of the FIELD: the GOAL-SIDE border (X = 0) and the FEEDER-SIDE border (X = 324 in.).
FLOATING BOARD	An elevated right-triangle plate, one per ALLIANCE, whose underside is 32 in. above the carpet, located at the FEEDER-SIDE border on the ALLIANCE's own half, against the CENTER LINE.
GOAL 1 / GOAL 2 / GOAL 3	HORIZONTAL GOALS with RIM heights of 18 / 38 / 60 in. Each ALLIANCE has one GOAL 1, two GOAL 2s and one GOAL 3.
GUARDRAIL	A 20 in. tall, 2 in. thick internal barrier that ROBOTS and CUBES cannot pass through (purple in figures).
HANG	End-of-MATCH ROBOT state defined in 5.3.3.
HANG RAIL	A 1.66 in. (42 mm) OD round pipe (1¼ in. Sch. 40) along the hypotenuse edge of each FLOATING BOARD, centreline 33 in. above the carpet.
HORIZONTAL GOAL	A goal whose opening (the THROAT) is horizontal and faces up. CUBES must enter from above, by being dropped or LAUNCHED.
INTRODUCTION POINT (IP)	A 12 × 12 in. opening at carpet level through which HUMAN PLAYERS introduce CUBES. There are 3 per ALLIANCE: IP-A, IP-B and IP-C.
LAUNCH	To shoot a CUBE into the air, kick it across the floor toward a desired location, or throw it forcefully (same meaning as 2025 G412).
LAUNCH ZONE	A 90 × 72 in. ZONE in front of an ALLIANCE's VERTICAL GOAL, which is at the opponent's end. It is taped in the colour of the ALLIANCE that uses it.

RELEASE LINE	The dashed ALLIANCE-coloured line 24 in. (610 mm) fieldward of a VERTICAL GOAL's opening plane. A legal VERTICAL GOAL shot must be released with the entire CUBE on the fieldward side of this line.
RIM	The top edge of a HORIZONTAL GOAL THROAT. RIM height is the scoring height of the goal.
ROOF	The pyramidal top of a HORIZONTAL GOAL outside the THROAT. It slopes down and away from the RIM at 40°, so CUBES that miss slide off.
SAFE ZONE	A 60 × 144 in. lane on each ALLIANCE's own end, bounded by the ALLIANCE WALL, the FEEDER-SIDE border, a GUARDRAIL and the ALLIANCE-coloured dashed tape. It is the only protected ZONE in SKYFORGE.
SCORE BIN	A container outside the FIELD that receives CUBES from a goal. Scored CUBES remain there until the MATCH ends.
STARTING ZONE	A 114 × 90 in. ZONE in the GOAL-SIDE corner of the ALLIANCE's own end (brown).
THROAT	The 11 × 11 in. (279 mm) square vertical passage below the RIM of a HORIZONTAL GOAL.
UNDER	End-of-MATCH ROBOT state defined in 5.3.3.
VERTICAL GOAL	A hexagonal opening in a vertical plane that faces the FIELD, mounted on top of the opponent ALLIANCE WALL. Its lowest point is 60 in. above the carpet.

4 ARENA & FIELD layout

4.1 FIELD overview

The FIELD is a 324 × 648 in. (27 ft 0 in. × 54 ft 0 in., 8.23 × 16.46 m) carpeted area bounded by two ALLIANCE WALLS and two FIELD BORDERS. Standard FRC field proportions are used rather than the sketch's proportions.


+X
RED ALLIANCE WALL
DS1
DS2
BLUE VERTICAL GOAL (Z ≥ 60 in.)
DS3
O
+Y
1
RELEASE LINE
RED STARTING ZONE
BLUE LAUNCH ZONE
IP-A
IP-B
IP-C
RED SAFE ZONE
RED FEEDER STATION (HUMAN PLAYERS, reserve CUBES)
2
3
RED FLOATING BOARD
CENTER LINE
BLUE FLOATING BOARD
BLUE SAFE ZONE
BLUE FEEDER STATION (HUMAN PLAYERS, reserve CUBES)
RED LAUNCH ZONE
BLUE STARTING ZONE
648 in. (54 ft 0 in. / 16.46 m)

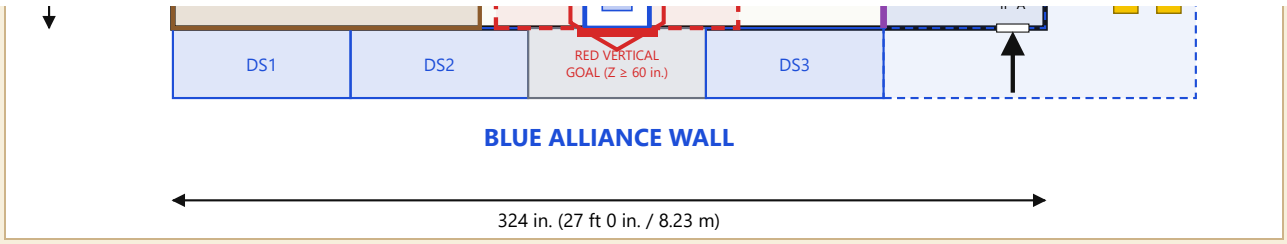


Figure 4-1 FIELD top view, to scale (1 drawing unit = 1 in.). Rectangles with numbers are HORIZONTAL GOALS: outline = 24 in. base, inner square = 11 in. THROAT. Numbers give the GOAL level and the colour gives the owner. Yellow squares are CUBES at staging positions. Dashed rectangles are LAUNCH ZONES, taped in the colour of the ALLIANCE that uses them. The hexagon symbol marks the VERTICAL GOAL, which is mounted on the ALLIANCE WALL above the opponent's GOAL 1.

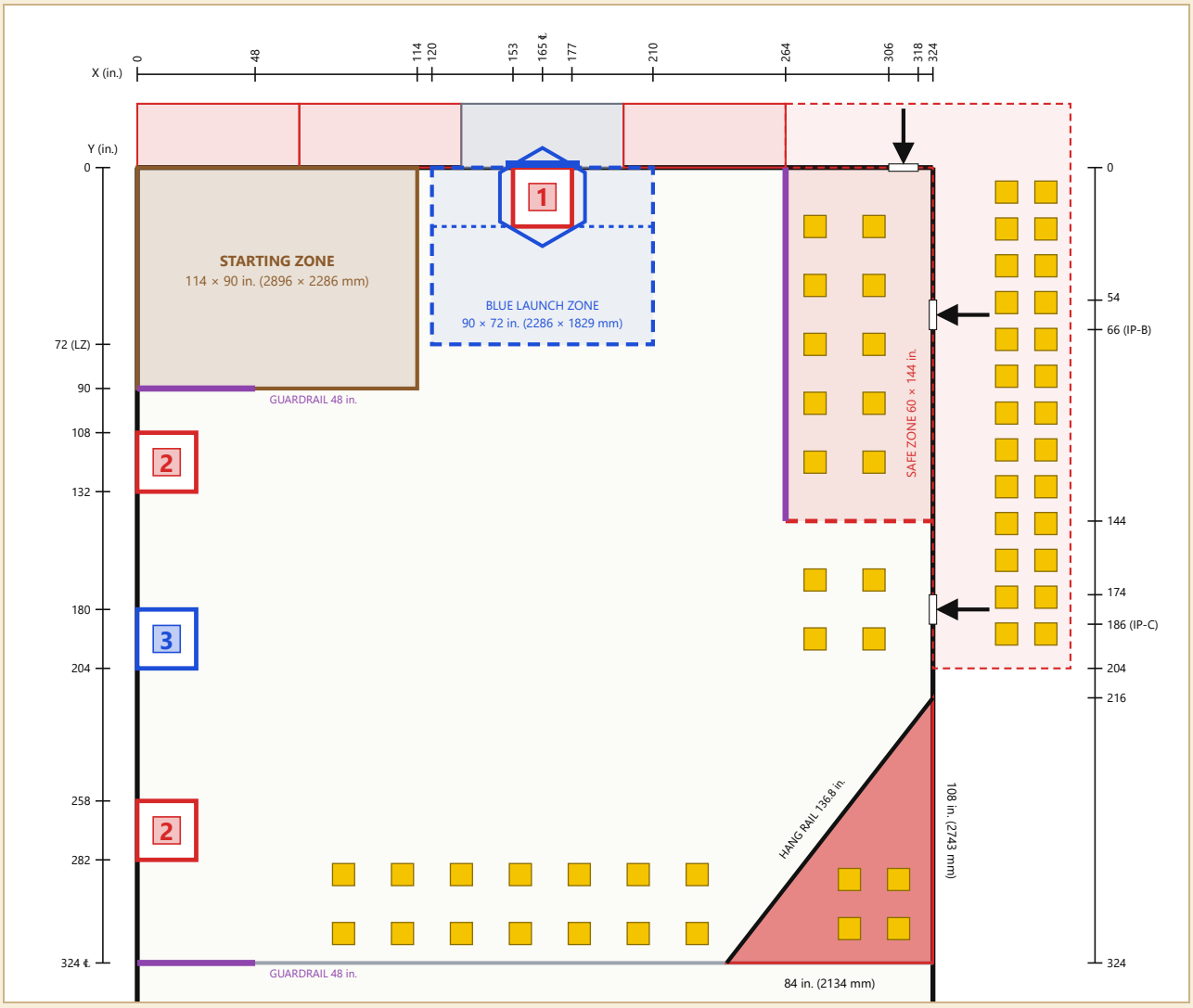


Figure 4-2 Red end, dimensioned. Rulers give absolute coordinates in inches from origin O. The blue end is the mirror image: $Y_{blue} = 648 - Y_{red}$.

4.2 FIELD BORDER and ALLIANCE WALLS

- **FIELD BORDER**: 20 in. (508 mm) tall clear polycarbonate on both long sides. The GOAL-SIDE border (X = 0) carries GOALS 2 and 3 and has openings behind them for their internal ramps. Each GOAL 1

discharges through its adjacent ALLIANCE WALL. The FEEDER-SIDE border (X = 324 in.) has IP-B and IP-C for each ALLIANCE.

- **ALLIANCE WALL** (per ALLIANCE, from X = 0):

Segment	X range (in.)	Height	Contents
DRIVER STATION 1	0 – 66	78 in. (1981 mm)	Behind the STARTING ZONE
DRIVER STATION 2	66 – 132	78 in.	
VERTICAL GOAL section	132 – 198	60 in. (1524 mm) solid wall, with the *opponent's* VERTICAL GOAL frame on top	Own GOAL 1 stands in front (Section 4.6)
DRIVER STATION 3	198 – 264	78 in.	
FEEDER corner	264 – 324	78 in.	IP-A opening (X 306 – 318); closes the SAFE ZONE

- DRIVER STATION structure, OPERATOR CONSOLE shelf, E-Stop/A-Stop and team signs are as in the 2025 ARENA.

4.3 GUARDRAILS

GUARDRAILS are 2 in. (51 mm) thick and 20 in. (508 mm) tall and are fixed to the carpet. Neither ROBOTS nor CUBES can pass through them.

GUARDRAIL	Red end	Blue end	Purpose
Starting guardrail	Y = 90, X 0 – 48	Y = 558, X 0 – 48	Separates the STARTING ZONE from the near GOAL 2; stops ROBOTS approaching the goal row along the wall from the corner
Center guardrail (shared)	Y = 324, X 0 – 48		Splits the goal row at the CENTER LINE; the two GOAL 2s nearest the CENTER LINE cannot be reached across it along the wall
SAFE ZONE guardrail	X = 264, Y 0 – 144	X = 264, Y 504 – 648	Forms the SAFE ZONE lane; the lane can only be entered through its dashed end (Y = 144 red / 504 blue)

4.4 ZONES

ZONE	Red	Blue	Tape	Notes
STARTING ZONE	X 0 – 114, Y 0 – 90	X 0 – 114, Y 558 – 648	Brown	Holds all 3 ROBOTS at MATCH start (G303)

LAUNCH ZONE	X 120 – 210, Y 576 – 648 *(at blue wall)*	X 120 – 210, Y 0 – 72 *(at red wall)*	Owner colour, dashed	Contains the opponent's GOAL 1. Not protected. Opponents inside it must stay at or below 48 in. (G416)
SAFE ZONE	X 264 – 324, Y 0 – 144	X 264 – 324, Y 504 – 648	Owner colour, dashed across the open end	Opponents may not enter (G420); ROBOTS inside are protected (G427)

4.5 HORIZONTAL GOALS

Each ALLIANCE owns four HORIZONTAL GOALS: one GOAL 1, two GOAL 2s and one GOAL 3. All four share the same construction and differ only in RIM height and location.

Item	Value
Base footprint	24 × 24 in. (610 × 610 mm), fixed to the carpet
THROAT (opening)	11 × 11 in. (279 × 279 mm) square, sides parallel to the FIELD axes, straight for 6 in. (152 mm) below the RIM
ROOF	Pyramidal, falling 5.5 in. (140 mm) from the RIM to the outer base edge (≈ 40°). CUBES that miss the THROAT slide off.
Clearance	1 in. (25 mm) per side for an aligned CUBE. A CUBE rotated about the vertical axis fits only within about ±14.8° ($9\cdot(\cos\theta + \sin\theta) \le 11$). A CUBE turned 45° measures 12.7 in. across its diagonal and cannot enter (Figure 4-3).
Delivery	From above only: dropped or LAUNCHED. The THROAT cannot be entered from the side.
After scoring	An internal ramp carries the CUBE into that goal's SCORE BIN outside the FIELD. GOALS 2 and 3 discharge through the GOAL-SIDE FIELD BORDER; GOAL 1 turns toward and discharges through a 12 × 12 in. (305 × 305 mm) opening in the adjacent ALLIANCE WALL directly behind its base. The wall opening is covered by the goal structure and is inaccessible to ROBOTS. Every scored CUBE stays in its bin until the end of the MATCH and cannot be de-scored.
Detection	Beam-break sensor plane 3 in. (76 mm) below the RIM, with REFEREE confirmation

Goal	RIM height	Red goals (centre X, Y)	Blue goals (centre X, Y)
GOAL 1 (low)	18 in. (457 mm)	(165, 12) – against the red wall, inside the blue LAUNCH ZONE	(165, 636) – against the blue wall, inside the red LAUNCH ZONE
GOAL 2 near (mid)	38 in. (965 mm)	(12, 120)	(12, 528)
GOAL 2 center (mid)	38 in. (965 mm)	(12, 270)	(12, 378)
GOAL 3 (high)	60 in. (1524 mm)	(12, 456) – *in the blue half*	(12, 192) – *in the red half*

Along the GOAL-SIDE border, from the red wall, the goals are: red 2 · blue 3 · red 2 | CENTER LINE | blue 2 · red 3 · blue 2. Each ALLIANCE's GOAL 1 and both GOAL 2s are in its own half. Its GOAL 3 and its VERTICAL GOAL are in the opponent's half.

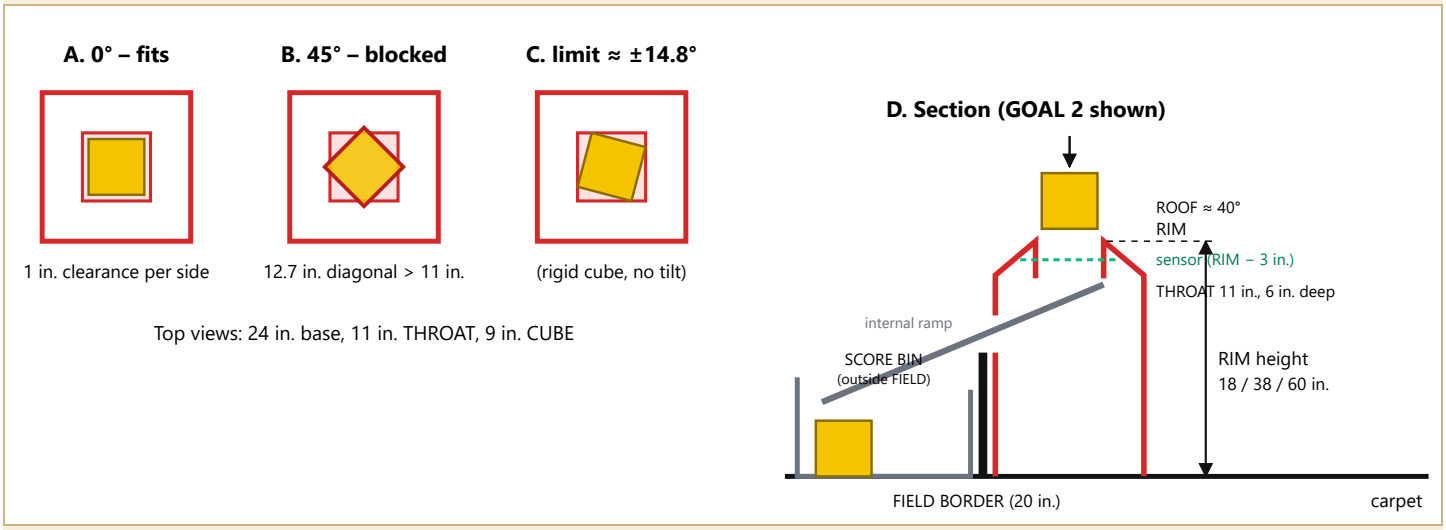


Figure 4-3 HORIZONTAL GOAL details. A–C: effect of CUBE yaw on entry. D: GOAL 2 section showing the RIM, THROAT, sensor plane, internal ramp and SCORE BIN. GOAL 1 has the same intake geometry at its 18 in. RIM, but its ramp turns through the adjacent ALLIANCE WALL as specified in 4.5.

4.6 VERTICAL GOALS and LAUNCH ZONES

- Each ALLIANCE has one VERTICAL GOAL, mounted on top of the 60 in. section of the **opponent's** ALLIANCE WALL. The blue VERTICAL GOAL is on the red wall, and the reverse.
- The opening is a regular hexagon with vertices at top and bottom, 30 in. (762 mm) across the flats (width) and 34.6 in. (880 mm) from vertex to vertex (height). Its centre is at X = 165 in. The lowest vertex is **60 in. (1524 mm)** above the carpet, the same height as the GOAL 3 RIM, and the top vertex is 94.6 in. (2403 mm). The opening plane is flush with the field-side face of the ALLIANCE WALL.
- Behind the opening, a capture net and chute hold scored CUBES out of the FIELD until the end of the MATCH. A sensor array in the chute counts them.
- The owner's **LAUNCH ZONE** (90 × 72 in.) is directly in front of it. The **opponent's GOAL 1** (24 × 24 in. base, X 153 – 177, Y 0 – 24 at the red wall) stands inside that LAUNCH ZONE, directly under the opening. A dashed **RELEASE LINE** spans the LAUNCH ZONE 24 in. (610 mm) fieldward of the opening plane, aligned with the fieldward edge of GOAL 1. G410 requires the delivering ROBOT's BUMPERS to intersect its LAUNCH ZONE, the whole CUBE to be fieldward of the RELEASE LINE when released, and the CUBE to travel unsupported through the opening. The blocker and RELEASE LINE therefore prevent adjacent placement.

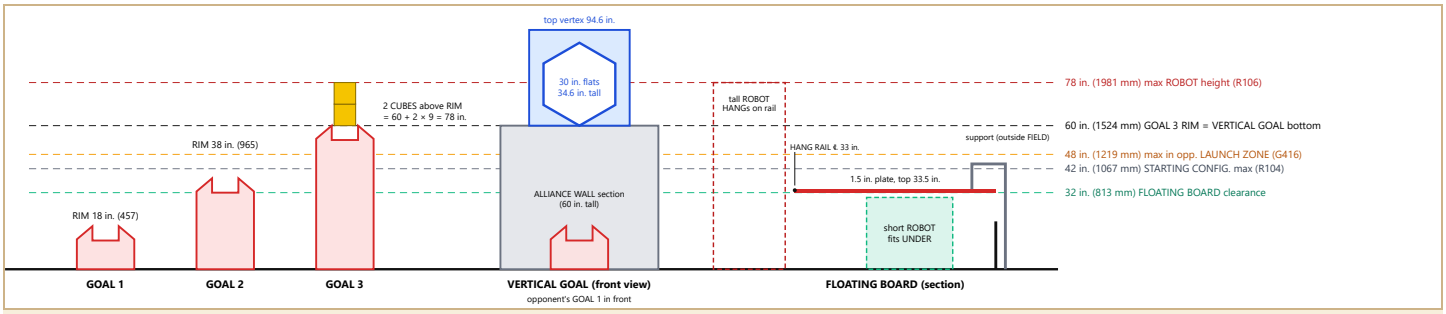


Figure 4-4 Height elevation (to scale vertically; elements placed side by side, not at their FIELD positions). All goal heights, the FLOATING BOARD clearance and the ROBOT height limits are shown on one scale.

4.7 FLOATING BOARDS

- Each ALLIANCE has one FLOATING BOARD on its own half. It is a right-triangle plate with legs of **108 in. (2743 mm)** along the FEEDER-SIDE border and **84 in. (2134 mm)** along the CENTER LINE. Red board vertices: (324, 216), (324, 324), (240, 324). Blue board: (324, 432), (324, 324), (240, 324). The two boards together form one larger triangle and meet at the CENTER LINE with a gap of 1 in. or less.
- The plate is 1.5 in. (38 mm) thick. Its underside is **32 in. (813 mm)** above the carpet, which is the minimum clearance under the board and rail, and its top is 33.5 in. (851 mm).
- A HANG RAIL (1.66 in. OD) runs the full 136.8 in. (3475 mm) of the hypotenuse, with its centreline 33 in. above the carpet. This is the only face of the board exposed to the FIELD.
- The boards are cantilevered from a steel frame outside the FEEDER-SIDE border. **There are no supports inside the FIELD**, so the area under each board is open carpet. The design load is at least 600 lb (272 kg) static per board.
- Largest square footprint that fits fully under one board: 84·108/(84+108) = **47.25 in. (1200 mm)**. A typical ROBOT is about 36–38 in. square including BUMPERS, so only **one** fits fully UNDER a board. Two ROBOTS side by side would each need to be 32.9 in. square or smaller.

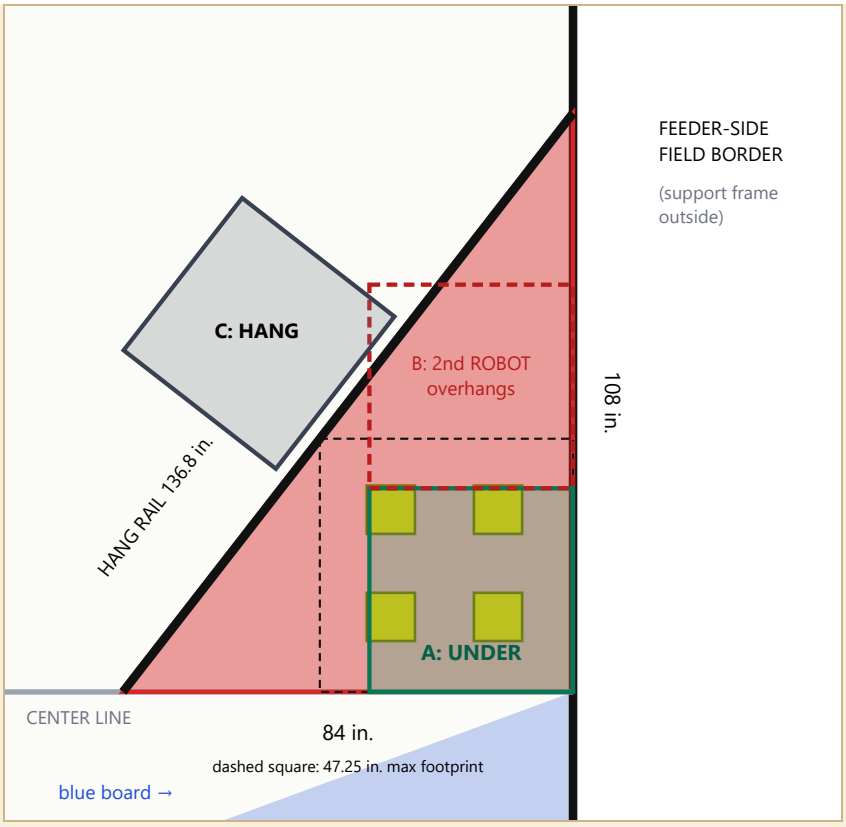


Figure 4-5 Red FLOATING BOARD, top view, to scale. A: one 38 × 38 in. ROBOT fits fully UNDER. B: a second ROBOT of that size overhangs the edge. C: a HANGING ROBOT on the HANG RAIL, clear of the UNDER position. The dashed square is the 47.25 in. theoretical maximum footprint.

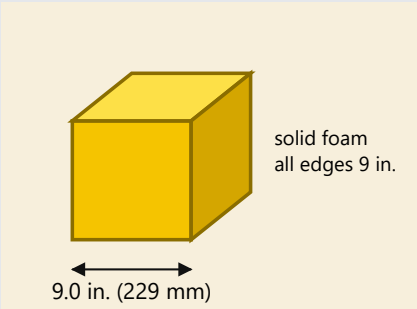
4.8 INTRODUCTION POINTS and FEEDER STATION

- Each INTRODUCTION POINT is a 12 × 12 in. (305 × 305 mm) opening at carpet level. On the outside it has a 24 in. long ramp at about 20°. A HUMAN PLAYER sets a CUBE on the ramp and releases it, and the CUBE slides onto the carpet.
- IPs are not used in AUTO (G404).

IP	Red location	Blue location	Enters	Protection
IP-A	ALLIANCE WALL, X 306 – 318	ALLIANCE WALL, X 306 – 318	SAFE ZONE (moving along the lane)	Protected
IP-B	FEEDER-SIDE border, Y 54 – 66	Y 582 – 594	SAFE ZONE (moving across the lane)	Protected
IP-C	FEEDER-SIDE border, Y 174 – 186	Y 462 – 474	Open FIELD, outside the SAFE ZONE	**Not protected**; opponents may collect these CUBES

The FEEDER STATION runs 204 in. along the FEEDER-SIDE border from the ALLIANCE WALL and wraps around the corner. Each ALLIANCE may station up to **2 HUMAN PLAYERS** there, taken from the DRIVE TEAM roles of 2025 Table 6-1.

4.9 SCORING ELEMENT: CUBE



Property	Value
Shape	Solid cube with lightly radiused edges (≤ 0.25 in.)
Edge length	9.0 in. ± 0.25 in. (229 ± 6 mm)
Material	Closed-cell polyethylene foam, yellow; the same concept as the FTC *RELIC RECOVERY* glyph (a foam cube of about 6 in.), scaled up 1.5×
Mass (estimate)	≈ 0.7 lb (≈ 325 g) at a nominal 1.7 lb/ft³ density; to be confirmed with a prototype
Why 9 in.	A legal ROBOT (≤ 120 in. perimeter, e.g. 30 × 30 in. frame) can reasonably carry 4 in a 2 × 2 or 1 × 4 arrangement, which matches the CONTROL limit of 4 (G409). Two CUBE heights (18 in.) are the headroom above GOAL 3 (R106).

4.10 CUBE staging

FIELD STAFF stage **224 CUBES** before each MATCH. Coordinates are CUBE centres for the red half; for blue, use Y → 648 − Y.

Location	Per ALLIANCE half	Total	Red-half positions (X; Y), in.
CENTER LINE rows	14	28	X = 84, 108, 132, 156, 180, 204, 228; Y = 288, 312 (blue: 336, 360)
SAFE ZONE lane	10	20	X = 276, 300; Y = 24, 48, 72, 96, 120 (a 19 in. corridor at X > 304.5 is left clear for IP-A and IP-B)
Beside IP-C	4	8	X = 276, 300; Y = 168, 192
Under FLOATING BOARD	4	8	X = 290, 310; Y = 290, 310
FEEDER STATION reserve (includes optional preloads)	80	160	Outside the FIELD. Each ROBOT may preload 1 taken from its ALLIANCE's reserve (G303-H).
Total	**112**	**224**	**64 on the FIELD at MATCH start (plus up to 6 preloaded)**

CUBES that leave the FIELD other than by scoring are returned by FIELD STAFF, when it is safe, to the FEEDER STATION of the ALLIANCE whose half they left from.

4.11 AprilTags

AprilTags follow the 2025 and 2026 FRC specification. There are 22 unique markers, IDs 1–22, from the 36h11 family. Each tag is 8.125 in. (20.64 cm) square including its white border, with a 6.5 in. (16.51 cm) black square, and is centred on a 10.5 in. (26.67 cm) square polycarbonate panel with an identifying text label. Worn tags are repaired with gaffer tape. As in 2026, several elements carry two tags (one centred and one offset), and the FLOATING BOARDS carry bracket-mounted tags, similar to the 2026 TRENCH.

ID red / blue	Location (on each ALLIANCE's end)	Centre X, Y (red), Z – in.	Faces
1 / 12	IP-A, on the ALLIANCE WALL above the opening	312, 0, 21.75	Down-field
2 / 13	IP-B, on the FEEDER-SIDE border beside the opening	324, 72, 10	Across the FIELD
3 / 14	IP-C, on the FEEDER-SIDE border beside the opening	324, 192, 10	Across the FIELD
4 / 15	ALLIANCE WALL below the opponent's VERTICAL GOAL, centred	165, 0, 44.25	Down-field
5 / 16	ALLIANCE WALL below the opponent's VERTICAL GOAL, offset	141, 0, 44.25	Down-field
6 / 17	GOAL 1, field face	165, 24, 6.5	Down-field
7 / 18	GOAL 2 near, field face	24, 120, 21.75	Across the FIELD
8 / 19	Opponent's GOAL 3 (in this half), field face	24, 192, 44.25	Across the FIELD
9 / 20	GOAL 2 centre, field face	24, 270, 21.75	Across the FIELD
10 / 21	FLOATING BOARD bracket, wall side (12 in. inboard of the HANG RAIL)	305.5, 259.4, 39.75	Toward the field, perpendicular to the HANG RAIL
11 / 22	FLOATING BOARD bracket, centre side	277.5, 295.4, 39.75	Toward the field, perpendicular to the HANG RAIL

Blue tag positions are the red positions mirrored across the CENTER LINE (Y → 648 − Y). The tag patterns are **not** mirrored, since a mirrored AprilTag does not decode. The same poses are published as a WPILib AprilTagFieldLayout JSON file with the field CAD. That file uses the WPILib convention: origin at the blue ALLIANCE WALL corner, +x down-field, +y toward the FEEDER-SIDE, +z up, in metres.

5 MATCH play & scoring

5.1 MATCH phases

Phase	Duration	Notes
AUTO	0:15	No DRIVE TEAM control. HUMAN PLAYERS may not introduce CUBES. Points are the same as in TELEOP.
(transition)	—	As in 2025, goal scoring is assessed for 3 s after the AUTO timer reaches 0:00, and those CUBES count as AUTO.
TELEOP	2:15	DRIVERS control ROBOTS. HUMAN PLAYERS may introduce CUBES through their IPs.
ENDGAME	last 0:20 of TELEOP	FLOATING BOARD protections apply (G419, G428).

5.2 Scoring assessment

Goal counts are updated throughout the MATCH and for up to 3 s after the timer reaches 0:00 at the end of AUTO and at the end of TELEOP. HANG and UNDER are assessed 3 s after the end of the MATCH, or when all ROBOTS have come to rest, whichever is first (2025 §6.5).

5.3 Scoring criteria

5.3.1 HORIZONTAL GOALS

A CUBE is scored in a HORIZONTAL GOAL once it has passed completely below the goal's sensor plane (RIM − 3 in.). It counts for the goal's owner, whichever ROBOT delivered it. CUBES resting on a ROOF, or stuck in the THROAT above the sensor plane, are not scored. Scored CUBES stay in the SCORE BIN and cannot be de-scored.

5.3.2 VERTICAL GOAL

A CUBE is scored in a VERTICAL GOAL only when a ROBOT of that goal's owning ALLIANCE makes a legal G410 delivery and the CUBE passes completely through the hexagonal opening into the capture chute. A CUBE delivered by the opponent or in violation of G410 is not scored, earns no MATCH points, and does not advance the SKYFORGE RP. FIELD STAFF remove such a CUBE from the capture chute after the MATCH.

5.3.3 ENDGAME states (assessed at the end of the MATCH)

- **HANG**: the ROBOT is in contact with its own ALLIANCE's FLOATING BOARD (including the HANG RAIL) and not in contact with the carpet. It may also contact only: CUBES; another ROBOT that qualifies for

HANG; a partner ROBOT contacted by an opponent in violation of G428; and opponent ROBOTS. (Adapted from the 2025 CAGE criteria.) Being supported on top of the board counts.

- **UNDER**: the vertical projection of the whole ROBOT, including BUMPERS, is entirely inside the vertical projection of its own ALLIANCE's FLOATING BOARD; the ROBOT does not qualify for HANG; and it is not in contact with the opponent's FLOATING BOARD.
- Each ROBOT scores at most one ENDGAME state.

5.4 Point values

Action	AUTO	TELEOP
CUBE scored in GOAL 1 (RIM 18 in.)	2	2
CUBE scored in GOAL 2 (RIM 38 in.)	3	3
CUBE scored in GOAL 3 (RIM 60 in.)	5	5
CUBE legally scored by its owning ALLIANCE in its VERTICAL GOAL (≥ 60 in.; G410)	4	4
ROBOT UNDER own FLOATING BOARD (end of MATCH)	—	3
ROBOT HANG on own FLOATING BOARD (end of MATCH)	—	10

There is no separate AUTO bonus. AUTO and TELEOP points are recorded separately only for tiebreakers.

5.5 Ranking Points (Qualification MATCHES)

RP	Requirement	Parameter (default)	RP
WIN	Finish the MATCH with more MATCH points than the opponent	—	3
TIE	Finish with the same number of MATCH points	—	1
LEVELS RP	At the end of the MATCH, the ALLIANCE's HORIZONTAL GOALS hold at least x_1 CUBES in GOAL 1, **and** at least x_2 CUBES in its two GOAL 2s combined, **and** at least x_3 CUBES in GOAL 3	$x_1 = 22$, $x_2 = 12$, $x_3 = 18$	1
SKYFORGE RP	At least y CUBES legally scored by the ALLIANCE in its own VERTICAL GOAL; opponent deliveries never contribute	$y = 30$	1
BOARD RP	At least 2 ROBOTS HANG, **or** at least 1 ROBOT HANGs and at least 1 other ROBOT is UNDER (on the ALLIANCE's own FLOATING BOARD)	—	1

The maximum is 6 RP per MATCH. As in 2025, the thresholds may be raised for District Championship and Championship events.

RP parameters (confirmed by the game designer). $x_1/x_2/x_3$ are per-level minimums in the HORIZONTAL GOALS (GOAL 1 / both GOAL 2s / GOAL 3), and y is the minimum in the ALLIANCE's VERTICAL GOAL. For the BOARD RP, "1 HANG + 1 UNDER" was chosen over "+ 2 UNDER" because only one ROBOT of typical size fits fully under a board (Section 4.7). These are harder draft scoring thresholds, not a promise of particular event success rates. See Section 10.4 for the saved-model evidence and its limitations.

5.6 Qualification ranking and playoff ties

Order	Qualification sort criteria (after the 2025 Table 10-1 structure)
1st	Ranking Score (average RP)
2nd	Average ALLIANCE MATCH points, not including MINOR and MAJOR FOULS
3rd	Average ALLIANCE VERTICAL GOAL points
4th	Average ALLIANCE ENDGAME (HANG + UNDER) points
5th	Average ALLIANCE AUTO points
6th	Random sorting by FMS

Playoff ties (a mock-specific order) are decided by, in turn: 1) more foul points awarded to the ALLIANCE (i.e. fouls committed by the opponent); 2) more AUTO points; 3) more ENDGAME points; 4) more VERTICAL GOAL points; 5) a replay.

6 Violations

These are unchanged from 2025 Table 6-3. Rules are called as perceived by a REFEREE, and each instance is assessed separately unless noted.

Penalty	Description
MINOR FOUL	2 points credited to the opponent's MATCH point total
MAJOR FOUL	6 points credited to the opponent's MATCH point total
YELLOW CARD	Issued by the Head REFEREE for egregious behaviour or rule violations. A second YELLOW CARD in the same tournament phase becomes a RED CARD.
RED CARD	The team is DISQUALIFIED for the MATCH
DISABLED	The ROBOT is commanded to deactivate all outputs for the rest of the MATCH
DISQUALIFIED	0 MATCH points and 0 RP in a Qualification MATCH, or 0 MATCH points for the ALLIANCE in a Playoff MATCH
VERBAL WARNING	A warning from event staff or the Head REFEREE
RP award / ineligibility	Some rules award an RP to the opponent (e.g. G419, G428) or make an ALLIANCE ineligible for one

YELLOW/RED CARD application, carry-over and Playoff ALLIANCE-wide CARDS follow 2025 §6.6.1–6.6.3. Violation wording such as “MAJOR FOUL, and for every 3 seconds…” is interpreted as in 2025 Table 6-5.

7 Game rules (G)

7.0 How to read this section

NEW SKYFORGE-specific rule MOD 2025 rule with changed content INH inherited: the 2025 rule of the same number applies with these term substitutions:

2025 term	SKYFORGE term
CORAL, ALGAE, SCORING ELEMENT	CUBE
REEF / BRANCH / trough, NET, PROCESSOR	HORIZONTAL GOAL / THROAT, VERTICAL GOAL, (none)
CAGE, BARGE, BARGE ZONE	FLOATING BOARD (incl. HANG RAIL)
CORAL STATION, CHUTE, PROCESSOR AREA	INTRODUCTION POINT, IP ramp, FEEDER STATION
ROBOT STARTING LINE	STARTING ZONE
REEF ZONE (protected)	SAFE ZONE (protected)

7.1 – 7.2 Safety and conduct (G1xx, G2xx)

INH G101–G104 (personal safety) and G201–G210 and G212 are inherited unchanged.

MOD **G211 Egregious or exceptional violations.** The 2025 rule is inherited except its item H is replaced by: “a ROBOT climbing on or attaching to the opponent's FLOATING BOARD, or any ROBOT or team member climbing on a FLOATING BOARD support structure.” A ROBOT climbing on, being supported by, hanging from, or attaching to its **own** FLOATING BOARD or HANG RAIL in compliance with G418 is normal SKYFORGE gameplay and is not a G211 violation.

7.3 Pre-MATCH

INH **G301 Be prompt. G302 You can't bring/use anything you want.**

MOD **G303 Start your ROBOTS.** A ROBOT must meet all of the following MATCH-start requirements: A. it does not pose a hazard; B. it has passed inspection; C. if modified since, it is compliant with I104; D. **its BUMPERS are fully inside its own STARTING ZONE**; E. it is the only team-provided item left on the FIELD; F. it is not attached to, entangled with or suspended from any FIELD element; G. it is in its STARTING CONFIGURATION (R102, R104); H. it fully and solely supports **no more than 1 CUBE**.

Violation: as 2025 G303.

7.4 In-MATCH

7.4.1 AUTO

INH **G401 Behind the lines. G402 Let the ROBOT do its thing** (the HUMAN PLAYER feeding exception does not apply, because there is no introduction in AUTO).

MOD **G403 Limited AUTO opponent interaction.** In AUTO, a ROBOT whose BUMPERS are completely across the CENTER LINE, in the opponent's half, may not contact an opponent ROBOT, either directly or through a CUBE CONTROLLED by either ROBOT, regardless of who initiates contact.

Violation: MAJOR FOUL and VERBAL WARNING. YELLOW CARD for subsequent violations during the event.

NEW **G404 No introduction in AUTO.** In AUTO, a HUMAN PLAYER may not introduce a CUBE onto the FIELD or place one on an IP ramp.

Violation: MINOR FOUL per CUBE.

NEW **G405 No opponent FLOATING BOARD in AUTO.** In AUTO, a ROBOT may not contact the opponent's FLOATING BOARD or HANG RAIL.

Violation: MAJOR FOUL.

7.4.2 CUBES

MOD **G406 ROBOTS: use CUBES as directed.** A ROBOT may not deliberately use a CUBE to ease or amplify a challenge associated with a FIELD element. Examples include LAUNCHING CUBES at ROBOTS; using CUBES to climb onto or stay on a FLOATING BOARD; wedging CUBES into or over an opponent THROAT; and stacking CUBES in front of an opponent IP.

Violation: MAJOR FOUL.

INH **G407 Keep CUBES in bounds** (scoring into a goal is not ejection). **G408 Don't abuse CUBES.**

MOD **G409 4 CUBES at a time.** A ROBOT may not simultaneously CONTROL more than **4** CUBES, directly or through other objects. Bulldozing and deflecting are not CONTROL, as in 2025.

Violation: MINOR FOUL per additional CUBE. YELLOW CARD if excessive, e.g. CONTROL of 6 or more, or greater-than-MOMENTARY CONTROL of 5 more than twice in a MATCH.

NEW **G410 Shoot your VERTICAL GOAL from behind the RELEASE LINE.** A ROBOT may deliver a CUBE only into its own ALLIANCE's VERTICAL GOAL, and only when all of the following are true at the instant the CUBE leaves the ROBOT: A. the ROBOT's BUMPERS intersect its own LAUNCH ZONE; B. the

entire CUBE is fieldward of the RELEASE LINE, at least 24 in. (610 mm) from the goal's opening plane; and C. after release, the CUBE is unsupported by any ROBOT or FIELD element until it passes through or misses the opening. A CUBE that rolls or bounces on the carpet, GOAL 1, ALLIANCE WALL, goal frame, or another ROBOT before reaching the opening is not a legal delivery.

Violation: MAJOR FOUL per CUBE. The CUBE is not scored and does not advance the SKYFORGE RP.

NEW **G411 No goal-tending.** A ROBOT may not A. break the vertical projection of an opponent THROAT above its RIM, or break the plane of that THROAT; B. break the plane of the opponent's VERTICAL GOAL opening or contact its frame; or C. contact an opponent's SCORE BIN or capture net.

Violation: MAJOR FOUL, and a further MAJOR FOUL for every 3 seconds the situation is not corrected.

NEW **G412 Scored CUBES stay scored.** No ROBOT or DRIVE TEAM member may remove a CUBE from any SCORE BIN or capture chute.

Violation: MAJOR FOUL per CUBE and YELLOW CARD. A CUBE that had already met its scoring criteria still counts.

7.4.3 ROBOT

INH **G413 ROBOTS must be safe. G414 Keep your BUMPERS low.**

MOD **G415 Expansion limits.** A ROBOT may not A. extend more than 18 in. (457 mm) beyond the vertical projection of its ROBOT PERIMETER, or B. extend more than **78 in. (1981 mm)** above the carpet. Over-expansion caused by damage and not used for strategic benefit is excepted, as in 2025.

Violation: MINOR FOUL, or MAJOR FOUL if used for strategic benefit, including if it impedes or enables a scoring action.

NEW **G416 Duck in the opponent's LAUNCH ZONE.** A ROBOT whose BUMPERS are partially or fully inside the *opponent's* LAUNCH ZONE may not extend more than 48 in. (1219 mm) above the carpet. This is based on 2019 G12 “Duck in the HAB ZONES”.

Violation: MINOR FOUL. MAJOR FOUL if the ROBOT blocks a CUBE LAUNCHED toward the opponent's VERTICAL GOAL.

INH **G417 Don't damage the FIELD** (2025 G416).

MOD **G418 Watch your FIELD interaction.** A ROBOT may not grab, grasp, attach to, become entangled with or suspend from any FIELD element **except its own ALLIANCE's FLOATING BOARD and HANG RAIL**. It may not contact any FLOATING BOARD support structure.

Violation: MAJOR FOUL, plus YELLOW CARD if REPEATED or longer than MOMENTARY. DISABLED if damage is likely.

NEW **G419 Opponent FLOATING BOARD is off-limits in ENDGAME.** During ENDGAME, a ROBOT may not contact the opponent's FLOATING BOARD or HANG RAIL, and its BUMPERS may not be partially or fully under the opponent's FLOATING BOARD.

Violation: MAJOR FOUL, and the opponent ALLIANCE is awarded the BOARD RP if it is a Qualification MATCH.

NEW **G420 Stay out of the opponent's SAFE ZONE.** A ROBOT's BUMPERS may not be partially or fully inside the opponent's SAFE ZONE.

Violation: MINOR FOUL, and a MAJOR FOUL for every 3 seconds the situation is not corrected. A MAJOR FOUL per CUBE CONTROLLED from inside it.

2025 G421 ("1 defender at a time") is **not** used. SKYFORGE requires every ALLIANCE to work in the opponent's half, because GOAL 3 and the VERTICAL GOAL are there.

7.4.4 Opponent interaction

INH **G422** Stay out of other ROBOTS. **G423** This isn't combat robotics. **G424** Don't tip or entangle. **G425** 3-count on PINS. **G426** Don't collude to shut down major parts of game play. Examples here include blocking all access to the opponent's GOAL 3 and VERTICAL GOAL together, or to all of their IPs.

MOD **G427 SAFE ZONE protection.** A ROBOT may not contact, directly or through a CUBE, an opponent ROBOT that is partially or fully inside that opponent's SAFE ZONE, regardless of who initiates contact.

Violation: MAJOR FOUL.

MOD **G428 FLOATING BOARD protection.** During ENDGAME, a ROBOT may not contact, directly or through a CUBE, an opponent ROBOT that is in contact with its own FLOATING BOARD or whose BUMPERS are partially or fully under it, regardless of who initiates contact.

Violation: MAJOR FOUL, and the opponent ALLIANCE is awarded the BOARD RP if it is a Qualification MATCH.

The LAUNCH ZONE is **not** protected, so defence there is legal within G416 and G422–G426. The opponent's GOAL 1 inside the zone is a deliberate obstacle.

7.4.5 Human

MOD **G429 No wandering.** As in 2025, except that HUMAN PLAYERS must stay in their ALLIANCE's FEEDER STATION.

Violation: MINOR FOUL.

INH **G430** Hands off the controls. **G432** Humans: use CUBES as directed. **G434** COACHES, CUBES are off-limits.

MOD **G431 Watch your reach.** A DRIVE TEAM member may not reach through an IP opening past the plane of the FIELD BORDER or ALLIANCE WALL.

Violation: MINOR FOUL.

MOD **G433 CUBE delivery.** CUBES may only be put onto the FIELD by a HUMAN PLAYER, during TELEOP, through one of their own ALLIANCE's IPs. At most one CUBE may be on each IP ramp at a time: the next CUBE may be placed only once the previous one is fully on the carpet.

Violation: MAJOR FOUL for entry by any other route. MINOR FOUL per extra CUBE on a ramp.

7.5 Post-MATCH

INH G501 Leave promptly.

8 ROBOT construction rules (R)

All 2025 R-rules apply unless changed below. That includes safety (R2xx), budget and fabrication (R3xx), BUMPERS (R4xx), motors and actuators (R5xx), power (R6xx), control, command and signals (R7xx), pneumatics (R8xx) and the OPERATOR CONSOLE (R9xx).

8.1 ROBOT size and expansion

Rule	Limit	Source
R101 ROBOT PERIMETER must be fixed	Non-articulating, within the BUMPER ZONE	INH 2025
R102 STARTING CONFIGURATION – no overhang	Nothing outside the vertical projection of the ROBOT PERIMETER except BUMPERS and minor protrusions	INH 2025
R103 Weight	≤ 115 lb (52 kg), excluding BUMPERS and battery	INH 2025
R104 STARTING CONFIGURATION – max size	ROBOT PERIMETER ≤ 120 in. (3048 mm); height ≤ **42 in.** (1067 mm, 3 ft 6 in.)	INH 2025 (2019 allowed 4 ft)
R105 Extension limit	≤ **18 in.** (457 mm) beyond the vertical projection of the ROBOT PERIMETER, in any direction and on any number of sides	INH 2025 (2019 allowed 30 in.)
R106 Height limit	≤ **78 in.** (1981 mm, 6 ft 6 in.) above the carpet **at all times**. This is the GOAL 3 RIM (60 in.) plus two 9 in. CUBES: a ROBOT can raise a CUBE over GOAL 3 but has only about two CUBE heights of headroom above it.	NEW
Area limit (game rule G416)	≤ 48 in. (1219 mm) while its BUMPERS are in the opponent's LAUNCH ZONE	NEW (2019 G12 concept)

As in 2025, teams must show at inspection that the ROBOT can constrain itself to R105 and R106 (with hardware or software) in every configuration.

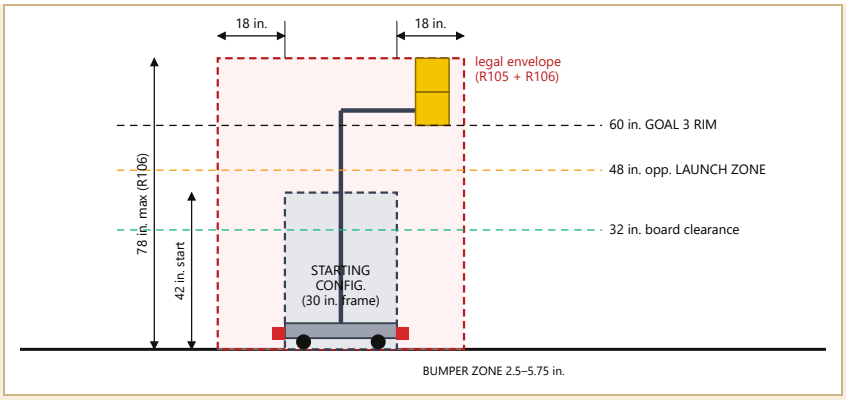


Figure 8-1 ROBOT envelope, side view, to scale, for an example 30 × 30 in. frame. The grey box is the STARTING CONFIGURATION (≤ 42 in. tall, within the ROBOT PERIMETER). The red dashed region is the envelope allowed during the MATCH: 18 in. beyond the perimeter and 78 in. total height. The example arm holds 2 CUBES above the GOAL 3 RIM height.

8.2 BUMPERS

INH 2025 R401–R412, including the BUMPER ZONE from 2.5 in. to 5.75 in. (63–146 mm) above the floor, padding of at least 2.25 in. depth, a 4 in. maximum extension from the ROBOT PERIMETER, and interchangeable red/blue covers with team numbers. A CUBE (9 in.) is taller than the top of the BUMPER ZONE, so ROBOTS cannot pass over CUBES and can herd them.

8.3 CUBE-specific construction note

INH R204 (leave SCORING ELEMENTS at the FIELD) and R206 (don't damage SCORING ELEMENTS) apply to the foam CUBES. Gripping surfaces that cut or permanently dent foam are likely to breach R206 and G408.

9 Inspection & tournament

INH 2025 Section 9 (Inspection & Eligibility), Section 10 (Tournaments; ranking criteria replaced as in 5.6) and Section 11 (Event rules). Inspection additionally checks R106 (78 in.) and the ROBOT's ability to meet G416 (48 in.) on demand.

10 Design notes, tuning parameters & open items

This section is not a rule. It records design intent and open questions so the game can be iterated.

10.1 Intended trade-offs

- **Go under or go tall.** A ROBOT that stays ≤ 32 in. can drive under both boards along the feeder side. It can also collect the 8 CUBES staged under the boards and score UNDER (3). It is naturally limited to GOAL 1 by placement (GOAL 1 needs a CUBE top of about 27 in.; GOAL 2 needs about 47 in.). A tall ROBOT reaches GOAL 2, GOAL 3 and the VERTICAL GOAL and can HANG (10), but has to go around the boards. A telescoping ROBOT that folds under 32 in. can have both, at the cost of mechanical complexity.
- **Near and far scoring.** GOAL 1 and both GOAL 2s are in the ALLIANCE's own half, near its IPs. GOAL 3 (5 pts) and the VERTICAL GOAL (4 pts, SKYFORGE RP) are in the opponent's half.
- **The blocker.** The opponent's GOAL 1 sits in your LAUNCH ZONE, directly under your VERTICAL GOAL. That stops ROBOTS driving up to the goal and gives the opponent's short ROBOTS a reason to be there. G416 lets them defend without being tall enough to block shots.
- **Precision over size.** The 11 in. THROAT for a 9 in. CUBE (±14.8° yaw) rewards controlled placement and makes LAUNCHING into HORIZONTAL GOALS possible but hard. The 30 in. hexagon is a larger target, but shots come from at least about 2 ft away.
- **Human-player risk.** IP-A and IP-B feed the protected SAFE ZONE. IP-C is faster to reach from mid-field, but opponents can take its CUBES.

10.2 Tuning parameters (defaults in this draft)

Parameter	Default	Affects
CUBE edge	9.0 in.	Capacity of 4, THROAT clearance, 78 in. height rule
THROAT size	11.0 in.	Placement tolerance; whether LAUNCHING into HORIZONTAL GOALS works
RIM heights G1 / G2 / G3	18 / 38 / 60 in.	Under-board ROBOT capability, height limit
Board clearance	32 in.	Main architecture trade-off
Board legs	108 × 84 in.	Number of ROBOTS fully UNDER (1)
VERTICAL GOAL size	30 in. across flats	Shooting difficulty
LAUNCH ZONE	90 × 72 in.	Shot distance and crowding
Points G1 / G2 / G3 / VG	2 / 3 / 5 / 4	Value of tall vs short ROBOTS
UNDER / HANG	3 / 10	Value of endgame and of going under

RP x_1, x_2, x_3, y	22, 12, 18, 30	Harder scoring RP; saved-model check in 10.4
CONTROL limit	4	Cycle size
Reserve CUBES per ALLIANCE	80	HP throughput vs FIELD supply

10.3 Open items and assumptions (not verified)

- No prototypes yet. Foam compression may change the real THROAT tolerance; the ±14.8° figure assumes a rigid cube with no tilt. Also untested: whether the 40° ROOF sheds foam CUBES, how CUBES bounce off the hexagon edges, the IP ramp exit speed and CUBE mass.
- RP thresholds and the reserve size are model-informed draft settings, not prototype or playtest results. The saved-match check in 10.4 does not model strategy adapting to the new thresholds or establish rates for a particular competition week. Point values are still estimates.
- The FLOATING BOARD design load (≥ 600 lb) and its cantilever frame have not been engineered.
- DRIVER STATION widths (66 in.) and FMS sensor details are placeholders. AprilTag poses are defined (4.11), but their visibility has not been checked with real cameras.
- Assumptions taken from the sketch: each triangle belongs to the ALLIANCE whose half it is in; each hexagon is the goal of the ALLIANCE whose colour it is drawn in, i.e. the one at the opponent's wall; standard 54 × 27 ft FIELD proportions replace the sketch proportions.

10.4 Balance revision and saved-model evidence

Retained rebalance. Compared with the original draft, LEVELS increases from 5 / 8 / 5 to **22 / 12 / 18** CUBES, SKYFORGE from 10 to **30**, and the FEEDER STATION reserve from 26 to **80 per ALLIANCE**, including optional preloads. The total supply increases from 116 to **224 CUBES**; the 64 loose starting CUBES and their locations are unchanged (4.10). This makes the scoring bonuses harder while giving HUMAN PLAYERS more supply to sustain cycling. BOARD criteria, point values and the CONTROL limit are unchanged.

Evidence: model, not measured play. The existing local SKYFORGE output `sim/out/event_population/matches.csv` contains 2,000 MATCHES (4,000 ALLIANCE-MATCHES). Its companion configuration uses seed 3 and a robot-tier mix of 0.5% world, 7.5% elite, 22% strong, 40% average and 30% rookie. The table below applies the current thresholds to those saved final goal and endgame counts. The saved RP flags do not consistently match the current thresholds, so they are not used as the new success rates. **No simulation was rerun for this revision.** These are fixed-trajectory checks, not predictions of how ROBOTS would change their strategy.

Bonus	Current requirement	Saved ALLIANCE-MATCHES passing	Rate
LEVELS	G1 ≥ 22, combined G2 ≥ 12, G3 ≥ 18	127 / 4,000	3.2%
SKYFORGE	VG ≥ 30	1,976 / 4,000	49.4%

BOARD	2 HANG, or 1 HANG + 1 other UNDER	2,864 / 4,000	71.6%

Historical context. The TBA-derived global weekly bonus rates illustrate why bonuses need not be equally easy: 2025 Coral was 5.6% in Week 1 and 19.9% in Week 6; 2024 Melody was 22.3% and 51.8%, respectively. The unchanged BOARD model rate is closer to 2025 Barge at DCMP (71.1%) than at regular events (38.7% in Week 1; 54.0% in Week 6). LEVELS is harder than these early-season examples and BOARD remains comparatively accessible. This supports a hard / medium / accessible bonus structure, **not** a claim that the mock population matches any particular week or event level. Rates also depend on alliance composition and winning versus losing.

CUBE availability. The saved event-population rows account for 77 reserve CUBES after three preloads per ALLIANCE, consistent with the 80-CUBE reserve. Reachable supply (reserve plus reachable loose and in-flight CUBES, not just the starting floor CUBES) was exhausted in 9 / 4,000 ALLIANCE-MATCHES (0.225%), first occurring 131–147 s into the 150 s MATCH, during ENDGAME. This supports retaining the larger reserve for that model population, but does not prove shortages are eliminated: saved strong-alliance studies still include late supply exhaustion. There is no matched before/after run here and no guarantee that 80 reserve CUBES suffice for every elite ALLIANCE.